8. $\int_{e}^{e^2} \frac{\ln x - 1}{x^2} dx$ 의 값은? [3점]

① $\frac{e+2}{e^2}$ ② $\frac{e+1}{e^2}$ ③ $\frac{1}{e}$ ④ $\frac{e-1}{e^2}$ ⑤ $\frac{e-2}{e^2}$